

The A-minor motifs in the decoding recognition process

A. Lescoute, E. Westhof*

*Institut de biologie moléculaire et cellulaire du CNRS,
UPR 'Architecture et Réactivité de l'ARN', université Louis-Pasteur, 15, rue R.-Descartes, 67084 Strasbourg, France*

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Abstract

The formation of A-minor motifs, mediated by adenines binding into the shallow/minor groove of stacked and helical Watson–Crick base pairs, is described. The conformations of the bacterial ribosomal decoding A site in various crystal structures are reviewed. The adenines A1492 and A1493 of the A site are seen either tucked in within the internal loop or bulging out and poised for interaction. This dynamic equilibrium contributes to the decoding process of the codon:anticodon base pairings. Aminoglycoside antibiotics lock the conformation of the A site in a single state with bulged-out adenines and thereby disrupt regulation of the decoding process.
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1. Introduction

The A site, located in the 30S subunit, is responsible for recognizing the aminoacyl tRNA bound with its anticodon to the in-frame codon of the messenger RNA. Its location was precisely and correctly described by chemical probing (see the paper by H. Noller in this issue) and later by X-ray crystallography [1]. It contains some of the most conserved nucleotides in ribosomes [2]. First, in order to describe the contacts made and the RNA motifs, we recall the nomenclature we use to annotate base–base interactions systematically and concisely [3].

2. Non-Watson–Crick base pairs

Broadly speaking, while Watson–Crick pairs define the secondary structure of an RNA, the non-Watson–Crick pairs are central to the formation of the tertiary structure. The architecture adopted by an RNA molecule is the result of the folding in space of the secondary structure elements maintained in precise positions by tertiary motifs in which non-Watson–Crick pairs

play a crucial role. It is therefore a necessity to be able to annotate them. To this end, any interacting assembly involving bases can be decomposed as a set of one-to-one base–base contacts. RNA purine and pyrimidine bases present three edges for hydrogen bonding interactions: the Watson–Crick edge, the Hoogsteen edge (for purine) or the C–H edge (for pyrimidine), and the Sugar edge (which includes the sugar 2'-hydroxyl group). A given edge of one base can potentially interact in a plane with any one of the three edges of a second base, and can do so in either the *cis* or *trans* orientation of the glycosidic bonds, i.e. with the attached sugars on the same side or on opposite sides of a line linking the interacting edges. This scheme leads to twelve possible, distinct edge-to-edge base-pairing geometries (or families). Each pairing geometry is designated by stating the interacting edges of the two bases (Watson–Crick represented by a circle, Hoogsteen represented by a square, or Sugar edge represented by a triangle) and the relative orientation of the glycosidic bonds, *cis* (filled symbols) or *trans* (empty symbols). A historically based priority rule is invoked for listing the bases in a pair: Watson–Crick edge > Hoogsteen edge > Sugar edge. Further details about the relative orientations of the sugar–phosphate backbones as a dependence of the base pairing type and on the families of isosteric base pairs can be found elsewhere [3].

For the family involving contacts between two sugar edges, several points are worth discussing. First, the triangle is

* Corresponding author.

E-mail addresses:

E.Westhof@ibmc.u-strasbg.fr, <http://www-ibmc.u-strasbg.fr/upr9002/westhof/> (E. Westhof).

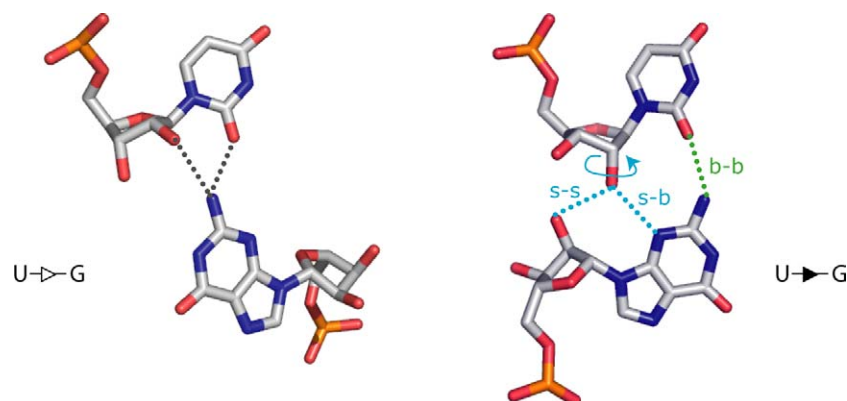


Fig. 1. Two examples of base pairs involving the sugar edges with the two possible orientations about the glycosidic bonds: a *trans* orientation (at the left), in which the sugars attached to the bases are on either side of a line drawn between the H-bonds connecting the two bases, and a *cis* orientation (at the right) in which the sugars attached to the bases are on the same side of a line drawn between the H-bonds connecting the two bases [3]. The symbols of the nomenclature are indicated (a triangle for the sugar edge, a square for the Hoogsteen edge and a circle for the Watson–Crick edge, filled with *cis* orientations and empty with a *trans* orientation). At the right, in green, a base/base contact is shown and, in blue, the sugar–sugar and the sugar–base contacts are shown. The curled arrow is intended to show that the hydrogen atom on the 2' hydroxyl group has the possibility to rotate around the C2'–O2' bond.

oriented so that the arrow points from the nucleotide giving the hydrogen atom of the 2'-hydroxyl group to the base of the paired nucleotide (see Fig. 1 for an example). Secondly, in a *cis* Sugar–Sugar pair, three types of interactions can occur: sugar–sugar (s–s) involving only the hydroxyl groups; sugar–base (s–b) involving one hydroxyl group and at least one base atom; base–base (b–b) involving only base atoms (see Fig. 1). For the UoG pair (right on Fig. 1), all three types occur. But, if the guanine is replaced by a pyrimidine, the single base–base contact would be lost. Within this family, interactions without any base–base H-bond belong to the ribose zipper interaction types [4,5]. In the absence of base–base contact, the relative positions of the interacting nucleotides are not constrained (because of the freely rotating hydrogen atom attached to the hydroxyl group). The A-minor interactions were first described from an analysis of the 50S ribosomal particle [6]. Four types of A-minor interactions were described but only the types I and II are restricted to adenines [6]. Here, only those latter two types will be further discussed. In type I, the adenine forms two 'pairing' contacts, a *cis* Sugar–Sugar and a *trans* Sugar–Sugar with the nucleotides of a Watson–Crick pair. In Type II,

the adenine forms a *cis* Sugar–Sugar pair with only one nucleotide of a Watson–Crick pair (Fig. 2 left and right). The A-minor interactions occur therefore between two nucleotides with at least one base–base H-bond.

3. The A-minor motif

An A-minor motif (type I/II) is the assembly of two consecutive adenines interacting in the shallow/minor groove of two consecutive Watson–Crick pairs through A-minor interactions (Fig. 3). In an A-minor motif, the 5'-A (A_n) makes the type II interaction and the 3'-A (A_{n+1}) the type I interaction. Thus, the second adenine, 3'-A (A_{n+1}), interacts with both strands of the receptor helix, while the first adenine, 5'-A (A_n), interacts only with one strand. Consequently, the second adenine (or 3'-A) forms more H-bonds with the Watson–Crick pair than the first one (or 5'-A). It is worth pointing out before discussing decoding that thermodynamic and mutations experiments have led to the following conclusions: (i) consecutive adenines recognize without any bias any two stacked Watson–Crick pairs and (ii) consecutive adenines have a strong preference

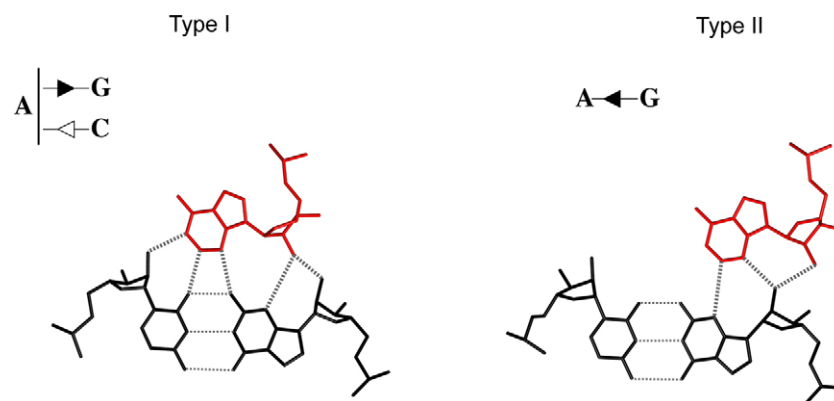


Fig. 2. Two examples of A-minor interactions, at the left the type I and at the right the type II (according to the designations of [6]). In type I, a C2-H...O2 contact is shown. In type II, the latter contact is replaced by a C2-H...N3 contact. The roles of C-H...O/N H-bonds or contacts in the stabilization of nucleic acids are now well documented.

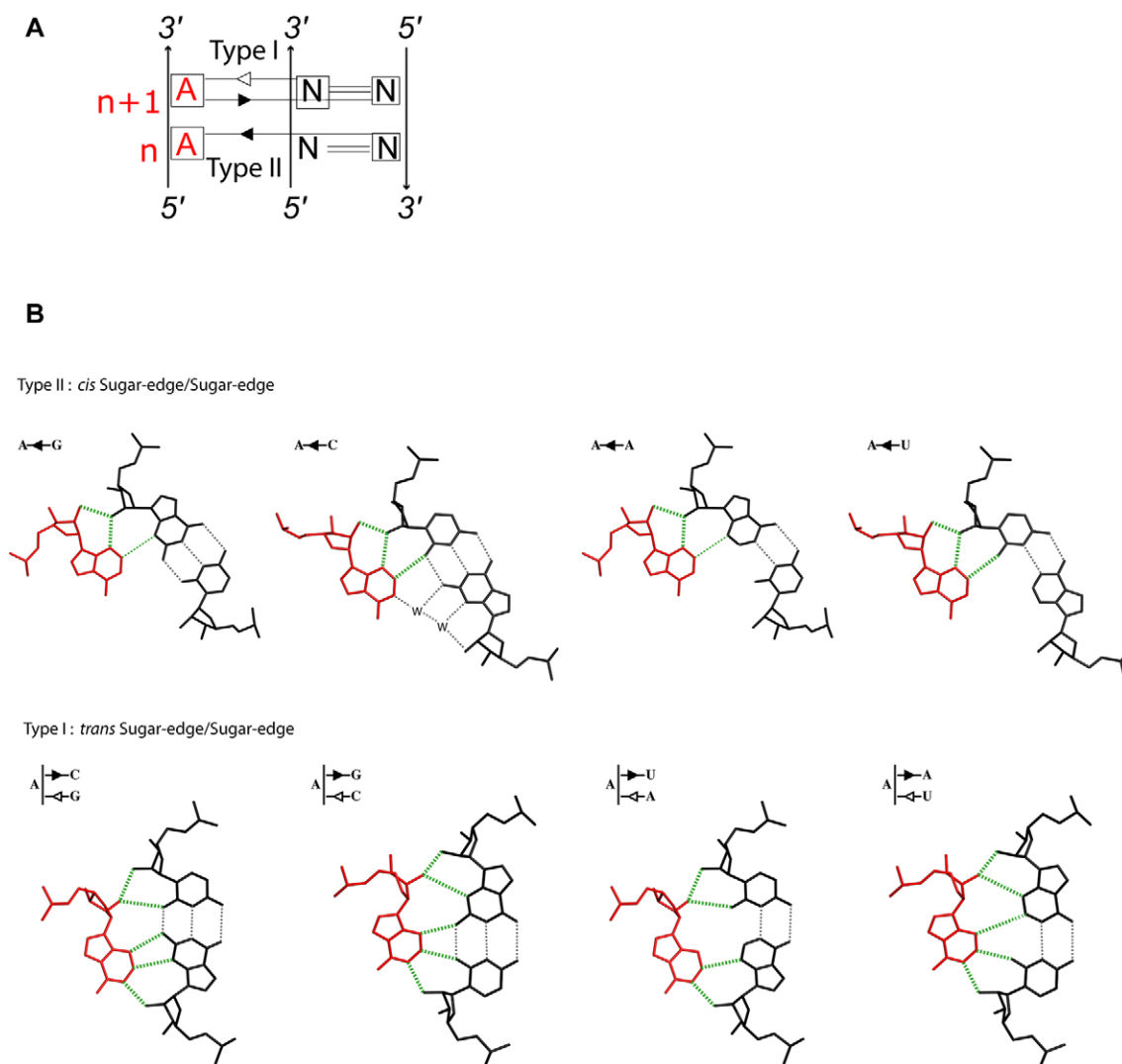


Fig. 3. (A) Two consecutive adenines forming successively a type II and a type I A minor interactions (in the 5' direction) form an A minor motif. (B) Theoretical representations of the binding to any successive Watson–Crick pairs.

for canonical Watson–Crick pairs compared to non canonical pairs, including “wobble” pairs [7,8].

After the crystal structure of the 50S particle appeared [9], it became clear that A-minor interactions constitute the most frequent tertiary RNA–RNA recognition motifs [10]. They are found not only in ribosomal RNAs but also in almost every ribozyme or structured RNA [11]. Their importance is consistent with the observation that, in secondary structures of rRNAs, adenines are very frequently seen in single stranded regions [12,13]. Interestingly, in the last sentence of one paper on chemical footprinting of 16S rRNA [14], Noller and co-workers remarked “Extensive protection of conserved, unpaired adenines upon formation of 30S subunits suggests that they play a special role in the assembly process, possibly providing signals for protein recognition”.

4. The crystal structures

In a series of remarkable crystal structures of 30S particles in the presence of various substrates, Ramakrishnan and co-

workers [1,15–18] demonstrated that this most common and frequent RNA–RNA interacting motif is at the very core of the recognition of the codon–anticodon helix and is central to the fidelity of the decoding process by the 30S subunit. Specifically, it was shown that A1492 and A1493 of the 16S A site recognize the minihelix formed between the anticodon bases and the triplet of the codon on the messenger RNA by forming an A-minor motif (Fig. 4). As in A-minor motif TypeI/TypeII, A1493, the 3' adenine, interacts with both the first base of the codon of the messenger and the third base of the anticodon (residue 36). The 5' adenine, A1492, interacts only with the second base of the codon of the messenger. The invariant residue G530 of the 16S rRNA binds to both A1492 and the second base of the anticodon (residue 35). It contacts also the sugar hydroxyl group of the third codon base (the wobble pair). The sterical fit and the number of H-bonds render the binding to complementary and canonical Watson–Crick pairs more favorable than to wobble or other non-canonical pairs. This tight fit between A-site nucleotides including G530 stabilizes therefore preferentially the minihelix formed between

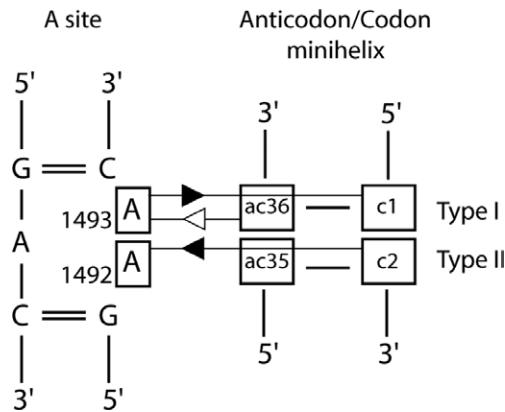


Fig. 4. In the decoding A site, A1492 and A1493 form an A-minor motif with the first two base pairs of the codon/anticodon helix resulting from the interactions between the tRNA anticodon and the messenger RNA codon [1].

cognate tRNAs and mRNA codons. With near-cognate tRNAs, the ‘open’ conformation with the bulging out of adenines is as or less stable compared to the ‘closed’ conformation with the tucked in adenines, leading to a conformationally dynamic A site which should lead to the release of tRNAs from the decoding ribosome. Experimental evidence had shown that (i) for binding to the A site, there is a slight preference for the messenger RNA to be a ribopolynucleotide instead of a deoxypolynucleotide [19]; (ii) in presence of aminoglycoside antibiotics (neomycin, streptomycin or kanamycin), deoxypolynucleotides can be translated by the ribosome [20] and finally (iii) that the only ribose requirement on the side of the tRNA was on residue U33 [21] which is key to the reversal of the chain required for forming a hairpin and the conformation of which is maintained by intramolecular H-bonds involving its hydroxyl group [22]. Both data are in agreement with the formation of an A-minor motif. The A-minor H-bonds to the 2'-OH of the mRNA nucleotides now explains those observations.

5. The A site is conformationally dynamic

The A-minor motif occurs in other RNA–RNA interactions. One motif, the GAAA [4,23], is extremely specific for a helical 11-nt receptor (Fig. 5) and the complex is thermodynamically very stable [24]. The GAAA tetraloop is a particular example among the frequently occurring tetraloops of the GNRA family [25]. Most generally, the GNRA tetraloops recognize two consecutive helical base pairs [26,27] (Fig. 5) by forming A-minor interactions. Other motifs are able to present two consecutive adenines for forming A-minor interactions (Fig. 5). In most instances, the two adenines are pre-organized for recognition and binding to their receptors by the local fold of which they are part. In GNRA tetraloops, the third base is stacked between the second and the fourth base which forms a *trans* Hoogsteen/Sugar base pair with the first residue [27,28]. In other types of motifs (Fig. 5), the adenines are involved in non-Watson–Crick pairs through their Hoogsteen edges. In this respect, the A site behaves rather differently. In empty 30S particles or in the structure of the *E. coli* ribosome, the conformations of the adenines in the internal loop of the A site are poorly defined and

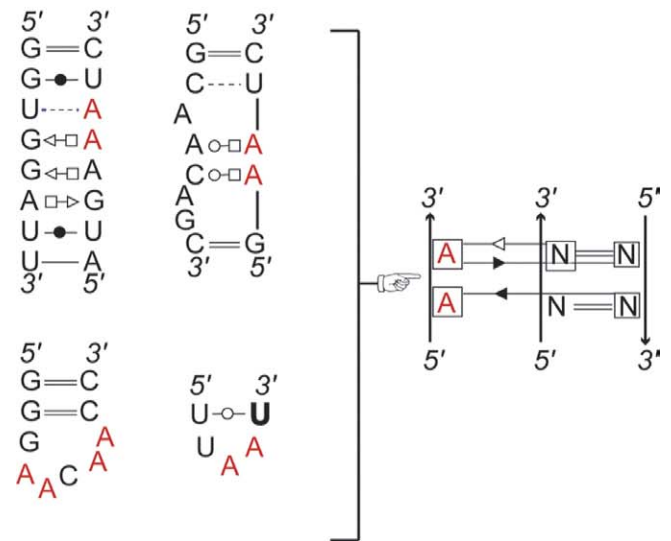
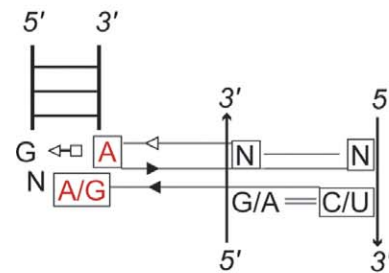
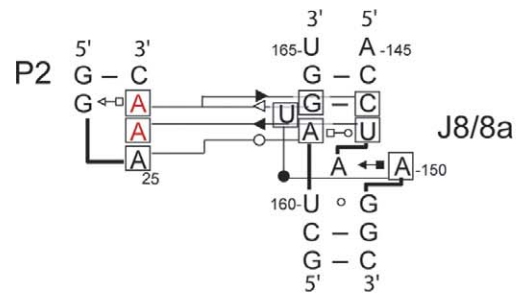


Fig. 5. Various RNA motifs with the possibility to form A-minor motifs: the GAAA tetraloop binds to a 11-nt receptor [23]; the GNRA tetraloop to two consecutive base pairs with restrictions on one of them [26]; other examples of motifs with non-Watson–Crick which bind to any consecutive helical pairs extracted from rRNA structures, PDB ID 1J5E and 1S72 [9,29,43]. The adenines which interact are colored red.

partly disordered [29,30]. Defined conformations, with two bulging adenines, are only observed in presence of the tRNA–mRNA complex or with aminoglycoside antibiotics [31]. The A site behaves therefore like a molecular switch, the dynamics of which is exploited for specific recognition of the codon:anticodon duplex. In most cases, the receptors of the A-minor motifs are constituted of stacked and helical Watson–Crick base pairs with a well-defined and stable geometry. However, NMR work has shown that the 11 nt-receptor for

the GAAA tetraloop is flexible in solution and adopts a different conformation by itself [32,33]. Thus, the known RNA–RNA assembly motifs based on A-minor interactions can occur between motifs with variable dynamics before association. The GNRA/Helix association occurs between two pre-formed motifs, the GAAA/11-nt motif occurs between two partially pre-formed motifs, with the A site displaying the highest dynamics difference between the unbound and bound states. Comparisons between structures of the free and antibiotic complexed A sites emphasize this truly astonishing dynamical system.

6. Aminoglycoside antibiotics lock the A site in a single conformation

Crystallographic structures of various 30S particles helped to visualize the mechanism of decoding as well as the action of aminoglycosides on the ribosome. During decoding, the A site changes its conformation from an “off” conformation (with A1492 and A1493 folded into the shallow groove of the A site) to an “on” conformation (with A1492 and A1493 fully bulged out from the A site) [1,15] (Fig. 6). The electron density was inconsistent with a single conformation for both adenines A1492 and A1493 in the absence of aminoglycosides, implicating a dynamic equilibrium of the internal loop [29]. Clear density was obtained after soaking an anticodon hairpin together with a single-stranded RNA into crystals of the 30S particle. The observed conformational change is necessary to allow A1492 and A1493 to interact specifically with the first two of the three base pairs formed by the cognate codon:anti-

codon interaction. It has been suggested that this structural change provokes the transition of the ribosome from an “open” to a “closed” form that is stabilized by contacts involving the cognate tRNA and the ribosome [16]. Interestingly, the structures of the minimal A-site RNA complexed to various aminoglycosides display crystal packings with intermolecular contacts between the bulging adenines (equivalent to A1492 and A1493) and the shallow/minor groove of a neighboring helix, mimicking those very contacts seen in the 30S co-crystals [34–38].

Thus, aminoglycosides lock the A site in a single open conformation and, as a consequence, the ribosome loses some of its ability to discriminate cognate versus non-cognate tRNA–mRNA associations and to transmit information about the decoding status to the large ribosomal subunit [16,17].

The crystal structures of a minimal A site complexed with several aminoglycosides characterized the binding mode of aminoglycosides at high resolution [34,35,38,39]. A particular interaction helps to maintain adenines 1492 and 1493 in the bulged-out conformation: the puckered sugar ring I of the antibiotic is inserted into the A-site helix by stacking against the guanine G1491 and by forming a pseudo-pair with two H-bonds to the Watson–Crick sites of the universally conserved adenine 1408. Thus, in the absence of a bound aminoglycoside in front of A1408 there would be a space void of nucleotide, an unstable situation which would explain the preference for a tucked in conformation (closed) of A1492 and A1493.

Indeed, in some structures, an empty A site was observed and several conformations were detected [38]. In one case, both A1492 and A1493 are tucked in, in a second structure

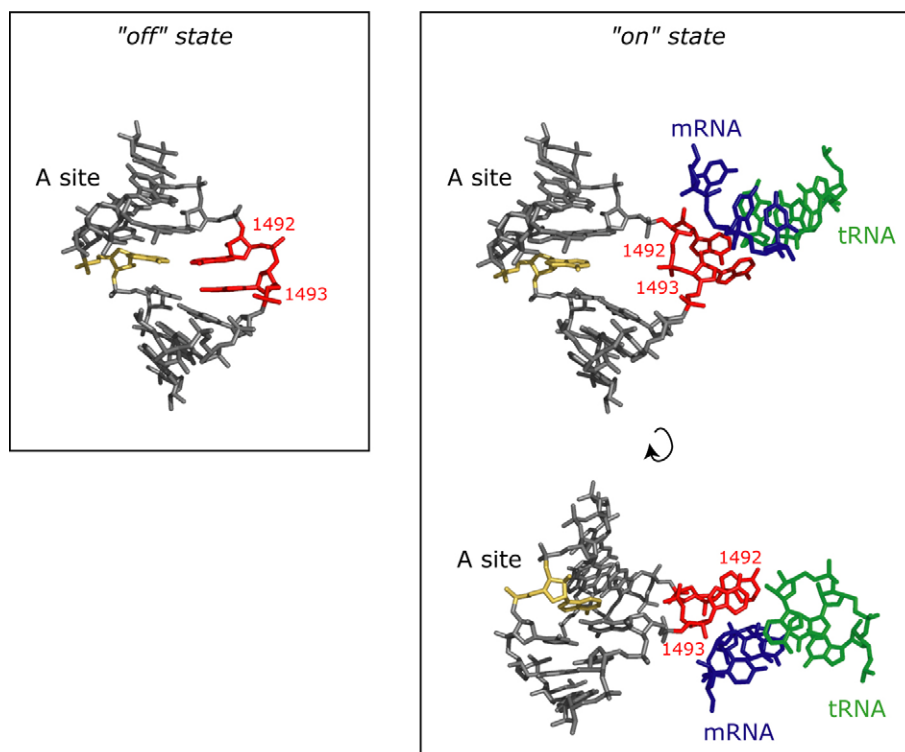


Fig. 6. The two main states of the A-site switch; the “off” state of 16S rRNA in the crystals of the 30S particles, PDB ID 1J5E [29] and two views of the “on” state in co-crystals containing 30S particles and fragments of the tRNA anticodon and mRNA, PDB ID 1IBM [1].

only a single adenine is bulged out (A1492) with the other, A1493, pairing to A1408, while a third one shows a reversal of roles for A1492 and A1493 [40]. A recent crystal structure (at 1.7 Å) of the empty A site inserted in a different oligonucleotide shows a two-state conformational disorder for one adenine: A1492 forms a *cis* Watson–Crick pair with A1408 (A1493 is bulged out) in one conformation, and both A1492 and A1493 bulge out and stack together in the other conformation [41]. The strongest evidence for a dynamic equilibrium influenced by aminoglycosides was obtained by correlating the latter crystal structure to fluorescence-induced changes of A-site fragments with the bulging adenines mutated to the fluorescent 2-aminopurine. The congruence between the fluorescence-induced changes and the crystal structures clearly indicated that paromomycin displaces A1492 from the internal loop, after which A1492 stacks with A1493 [41]. On the dynamic aspects of the A site see also the articles by Kevin Sanbonmatsu in this issue.

7. Evolutionary considerations

The central role of A-minor motifs in the decoding process bears on the RNA world hypothesis [2,42]. First, it is striking that the most common RNA–RNA recognition motif is used for the most crucial moment in protein synthesis, fidelity of decoding. The A-minor motifs monitor unspecifically the width of the shallow/minor groove of any RNA helical arrangement. Because they can form with any consecutive and stacked Watson–Crick base pairs, A-minor motifs are mutationally robust. In other words, A-minor motifs can accommodate many combinations of neutral mutations. Secondly, A site motifs are not frequent in other structured RNAs. It is not clear whether this is due to the extreme mobility of the free A site. It is not clear either what is the contribution of this mobility to the mechanism of the decoding process. Finally, around the A site only 16S rRNA nucleotides are found and implicated in the recognition (only some amino acids of S12 have a tangential participation). Those observations on the decoding process, together with the conclusion that “the ribosome is a ribozyme” [9,42], provide evidence that the ribosome evolved from a purely RNA based assembly. Clearly, once such a recognition system is so centrally incorporated in the decoding process, it cannot be replaced easily but can only be more finely tuned in the overall fidelity process.

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